ENGINEERING PORTFOLIO

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Selected R&D Projects in Embedded Systems, RF Hardware, and Marine Telemetry
GitHub Portfolio: handiko.github.io

This dossier outlines core technical engineering projects, hardware layouts, firmware implementations, and practical methodologies cultivated over 5+ years of production engineering experience across maritime, agrotech, and telecommunication sectors.

CORE TECHNICAL PROJECTS

1. IoT-Enabled AIS Receiver & Gateway

Tags: Production Level R&D | STM32F4 | ESP32 | NMEA-0183 | Embedded C++

Project Overview & Architecture

An enterprise-grade Internet-of-Things automatic monitoring gateway built to track commercial and local marine vessel networks. The system captures over-the-air raw maritime RF telemetry signals and parses them into deterministic data streams for transmission to centralized cloud services.

Key Technical Implementations

- Designed a high-throughput dual-MCU hardware bridge combining the raw computing capability of an ARM Cortex-M4 (STM32F4) with an ESP32 for high-bandwidth wireless edge connectivity.
- Implemented localized parsing firmware capable of transforming asynchronous RF demodulated frames into robust compliant NMEA-0183 data packets under high channel noise conditions.
- Built an integrated asynchronous HTTP configuration web server executing natively on-chip to handle field tuning, edge diagnostics, and secure network provisioning.

2. Low-Cost AIS Class-B Transmitter for Small Vessels

Tags: Proprietary Innovation | RF Design | Low Power Architecture | Supply-Chain Resilient

Project Overview & Architecture

An affordable, regulation-compliant marine telecommunication tracking device targeted at small-scale local boat fleets. This engineering initiative radically lowered commercial entry barriers for local operators by utilizing commercial off-the-shelf (COTS) components optimized for highly challenging marine environments and tight RF spectrum constraints.

Key Technical Implementations

- Engineered a highly resilient VHF radio-frequency layout optimized for low harmonic distortion and precise impedance matching across active maritime spectrum bands.
- Designed custom power distribution circuits coupled with fine-grained low-power microcontroller sleep

profiles to optimize operation longevity from standard boat marine batteries.
- Bypassed specialized high-cost telemetry microchips in favor of globally abundant, cost-effective microcontrollers to ensure manufacturing supply chain resilience.

3. Arduino APRS (Automatic Packet Reporting System) Tracker

Tags: Open Hardware & Communications | APRS Protocol | GPS Integration | Custom Shield Layout

Project Overview & Architecture

A compact telemetry extension hardware expansion shield enabling automatic radio-link-based geospatial tracking. Widely used across the amateur radio infrastructure for decentralized grid position updates without dependency on cellular or internet networks.

Key Technical Implementations

- Developed a production-ready custom PCB extension shield tracking active high-sensitivity GPS modules, baseband audio modulation sub-circuits, and custom radio interfaces.
- Authored low-overhead Audio Frequency Shift Keying (AFSK) modulation firmware to cleanly encode AX.25 UI telemetry packets through conventional handheld VHF radio architectures.